

Copilot Readiness Analyzer for Power BI

Product Requirements Document — Developer Side (Phase 1 & 2)

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1. Problem Statement

Power BI Copilot requires reports and semantic models to meet a baseline of structural and metadata quality to function effectively. However, the majority of existing reports in production were built before AI features existed — with no consideration for Copilot compatibility. Inconsistent measure naming, missing descriptions, poorly structured calculations, and thin metadata are the norm, not the exception.

For a regional organization with approximately 300 reports across 3 companies, mid-level and senior analysts face a compounding challenge: they are the primary report developers, often inheriting reports they did not build, already operating under heavy workload, and have no clear guidance or tooling to assess which reports are AI-ready or what specifically needs to change.

The result is that when Copilot is used on these reports today, it produces vague, unreliable, or incorrect outputs — silently eroding trust in AI features before they ever get a fair chance.

The Gap: There is no structured way for a developer to know *what is wrong*, *how bad it is*, or *where to start* — making AI adoption feel overwhelming rather than incremental.

2. Opportunity Sizing

Internal — Organization Level

The organization has approximately 300 Power BI reports across 3 companies, of which 200 are actively used. With a team of 7 analysts already operating at capacity, manually auditing each active report for Copilot compatibility would require an estimated 4 hours per report — totalling approximately 800 hours of remediation work before any AI value is unlocked.

200

Active reports requiring audit

800 hrs

Manual remediation effort estimated

7

Analysts on team already at capacity

The compounding cost: 800 hours is the equivalent of one analyst's entire working year — doing nothing but cleanup. Every month this goes unresolved, Copilot continues to underperform on live reports, damaging confidence in AI features and slowing broader AI adoption across the organization.

External — Market Level

This is not an isolated problem. Most enterprise Power BI environments share the same condition — large report libraries built before AI features existed, with no migration path provided by Microsoft. Organizations globally are investing in Copilot licences but seeing poor returns because the underlying reports are not ready.

The addressable market includes any mid-to-large organization running Power BI at scale — a significant segment of Microsoft's Fabric and Copilot customer base. A structured AI readiness solution represents a clear gap in the current tooling ecosystem.

3. Proposed Solution

Product Name (working title): Copilot Readiness Analyzer for Power BI

A developer-facing tool that scans an organization's Power BI semantic models and reports, evaluates their compatibility with Copilot, and gives analysts a clear, prioritized path to making their reports AI-ready.

Phase 1 — Audit & Recommend (MVP)

When a developer opens the tool, they land on a **Report Readiness List** — all active reports ranked from least to most AI-ready, so they know exactly where to focus first without reviewing 200 reports manually.

Each report receives an AI Readiness Score (0–100) across four dimensions:

Dimension	What is evaluated
Measure Naming Conventions	Are measures named clearly and consistently in a way Copilot can interpret?
Missing Descriptions	Do measures and columns have metadata descriptions that ground Copilot's understanding?
Calculation Structure & Complexity	Are DAX calculations structured legibly, or are they deeply nested and opaque?
Semantic Model Metadata Completeness	Is the broader model sufficiently documented for Copilot to reason over it?

Phase 2 — Auto-Remediation (Follow-on Release)

Once the audit layer is validated and trusted, the tool gains the ability to act — not just advise. Every auto-fix goes through a developer review and approve step before being applied. No change is ever made silently.

- **Rename measures:** Apply naming conventions automatically
- **Auto-generate descriptions:** LLM-generated metadata surfaced for review before applying
- **Suggest restructured calculations:** Flag complex DAX with restructuring suggestions, developer approves
- **Column renaming:** Plain-language alternatives proposed; originals preserved as hidden synonyms
- **Flag dormant assets:** Unused measures, fields, relationships, and report pages flagged for cleanup

What this is NOT: The tool does not modify reports without developer approval. It does not replace developer judgment — it amplifies it. Phase 1 ships as read-only. Phase 2 changes always require explicit approval before being applied.

4. User Stories

US-01 Report Prioritization

Phase 1 — MVP

As a senior analyst, I want to see all reports ranked by AI readiness score so that I can decide where to focus remediation effort first without manually reviewing 200 reports.

Acceptance Criteria: Reports list loads with scores visible, sortable, and filterable by company and workspace.

US-02 Report Audit Drill-Down

Phase 1 — MVP

As a BI developer, I want to open any report and see a detailed breakdown of every Copilot compatibility issue so that I know exactly what to fix and why it matters.

Acceptance Criteria: Each issue is categorized by dimension, assigned a severity (critical / warning / suggestion), and accompanied by a specific recommended action.

US-03 Inherited Report Triage

Phase 1 — MVP

As a analyst who has inherited reports I did not build, I want the tool to explain each issue in plain language so that I can remediate reports I am not deeply familiar with.

Acceptance Criteria: Every recommendation uses plain English — no assumption of prior knowledge of the report's structure.

US-04 Progress Tracking

Phase 1 — MVP

As a team lead, I want to track how our overall AI readiness score improves over time so that I can report progress to leadership and keep the team motivated.

Acceptance Criteria: A readiness trend view shows score changes per report and organization-wide over time.

US-05 Description Generation with Uncertainty Approval

Phase 2 — Auto-Remediation

As a BI developer, I want the tool to auto-generate descriptions for my measures and columns so that I do not have to write metadata manually for hundreds of fields.

Acceptance Criteria: Generated descriptions are surfaced for review before applying. Where the tool cannot confidently infer what a measure does, it flags it as 'Needs Your Input' and prompts the developer with a plain-language question before generating.

US-06 Dormant Asset Cleanup

Phase 2 — Auto-Remediation

As a senior analyst, I want the tool to flag all unused assets in my semantic model so that I can reduce the noise that causes Copilot to return irrelevant answers.

Acceptance Criteria: *Flagged assets include unused fields, measures, orphaned relationships, and empty report pages — each with last-used date and a one-click mark-for-removal option pending developer confirmation.*

US-07 Column Renaming for Copilot Clarity

Phase 2 — Auto-Remediation

As a BI developer, I want the tool to identify technical or ambiguous column names and suggest human-readable alternatives so that Copilot can accurately interpret and reference them.

Acceptance Criteria: *The tool proposes plain-language alternatives with reasoning. Developer accepts, edits, or dismisses each suggestion. Original names are preserved as hidden synonyms to avoid breaking existing report references.*

5. AI-Specific Considerations

This section addresses what can go wrong because this is an AI product, and how the design accounts for it.

1. Hallucination Risk — Wrong Descriptions

The tool will occasionally generate measure descriptions that are plausible but incorrect, particularly for complex or ambiguous DAX logic. Incorrect descriptions are shown normally but logged silently for quality review. A confidence score is computed per description — descriptions below a defined threshold are flagged internally and reviewed periodically to improve generation quality over time. This avoids interrupting developer workflow while building a feedback mechanism that improves accuracy over time.

2. Scoring Uncertainty — Measures the Tool Does Not Fully Understand

Measures containing logic too complex or context-dependent for confident evaluation are scored conservatively by default — assumed low readiness until verified. They are clearly marked as 'Pending Verification' in the audit view, with a prompt for the developer to provide context or confirm intent. Once verified, the tool re-scores with the additional context and updates the report's overall AI Readiness Score. This ensures scores are never falsely inflated, while giving developers a path to improve them through their own knowledge.

3. Trust & Developer Control — Disagreeing with Recommendations

Developers who know their reports may legitimately disagree with a recommendation. Developers can dismiss any recommendation but must provide a reason before dismissal is confirmed. Reasons are logged to identify patterns where the tool is systematically wrong, improve future recommendation accuracy, and surface recurring disagreements for model refinement. Dismissal reasons are not reviewed by team leads — they feed directly into the improvement loop, keeping the experience non-punitive.

4. Human-in-the-Loop — Non-Negotiable for Phase 2

Every auto-fix in Phase 2 requires explicit developer approval before being applied. No change is ever made silently. For column renames, original names are preserved as hidden synonyms to prevent breaking existing report references, visuals, or DAX dependencies. The tool's role in Phase 2 is to prepare the action, not execute it. The developer executes.

5. Confidence Transparency

The tool communicates its own uncertainty clearly throughout: readiness scores on unverified measures are visually distinct from verified ones; generated descriptions carry a confidence indicator; recommendations include a plain-language explanation of why the suggestion is being made, so developers can make informed decisions rather than blindly accepting AI output.

Design philosophy: The tool improves through use, but never acts without permission. In an enterprise context, trust is slow to build and fast to lose.

6. Success Metrics

North Star Metric: Average AI Readiness Score across all active reports — measured monthly. This single number captures whether the tool is achieving its core purpose: moving the organization's report estate from Copilot-incompatible to Copilot-ready over time.

Adoption Metrics

Metric	Target	Timeframe
% of active reports audited	80% of 200 active reports	First 30 days
Recommendations acted on	60% acceptance rate	First 60 days

Effectiveness Metrics

Metric	Target	Timeframe
Average AI Readiness Score improvement	+20 points across active reports	90 days post-launch
Copilot query quality improvement	Baseline Month 1; +30% by Month 3	90 days post-launch

Trust & Quality Guardrail Metrics

These metrics set the floor — they tell us if something is going wrong.

Metric	Guardrail	Action if Breached
Description dismissal rate	No more than 30%	Review LLM generation quality
'Needs verification' measure rate	No more than 25%	Improve DAX parsing capability
Developer trust score (quarterly survey)	4 / 5 or above	Conduct developer interviews

What we are NOT measuring: Speed of remediation (we do not want developers rushing fixes to hit a number) and number of auto-fixes applied in Phase 2 (quality over volume, every time).

7. Launch Checklist

Phase 1 — Internal Launch (7 Analysts, 3 Companies)

Product Readiness

- AI Readiness Score validated against at least 20 real reports before launch — reviewed by 2+ analysts [Phase 1]
- Audit drill-down view tested across all four scoring dimensions [Phase 1]
- 'Pending Verification' flow tested end-to-end — developer input updates score correctly [Phase 1]
- Dismissal reason capture working and logging correctly [Phase 1]
- Read-only mode confirmed — no write access to reports in Phase 1 [Phase 1]
- Tool tested against both owned and inherited reports [Phase 1]

Data & Access

- Power BI REST API connection established and authenticated across all 3 company workspaces [Phase 1]
- Report inventory confirmed — 200 active reports identified and loaded [Phase 1]
- Dormant reports (100) excluded from primary view but accessible [Phase 1]
- Data access permissions reviewed — tool reads semantic model metadata only, no underlying data [Phase 1]

Trust & Communication

- All 7 analysts briefed on what the tool does and does not do before launch [Phase 1]
- Plain-language explainer document prepared for each recommendation type [Phase 1]
- Feedback channel created beyond the in-product dismissal flow [Phase 1]
- Team lead dashboard live — org-wide readiness score visible from day one [Phase 1]

Baseline Established

- Month 1 Copilot query quality baseline captured — developers rate answer quality on 10 reports [Phase 1]
- Current average AI Readiness Score recorded as starting benchmark [Phase 1]
- Recommendation acceptance rate tracking live from day one [Phase 1]

Phase 1 → External Launch Gate

Before pitching to other Power BI shops, the following must be true:

- 80% of active reports audited within 30 days of internal launch
- Average AI Readiness Score improved by at least +15 points across the report estate
- Developer trust score 4/5 or above in first internal survey
- At least 3 analyst testimonials or quotes captured for external pitch
- One documented case study ready — before/after score, time saved, Copilot improvement observed

Known Risks Before Launch

Risk	Mitigation	Severity
LLM generates poor descriptions for complex DAX	Silent logging + confidence threshold in place from day one	Medium
Analysts do not trust the score	Transparent scoring breakdown per dimension — no black box	High
API access restricted by IT across some workspaces	IT sign-off secured pre-launch for all 3 companies	High
Analysts too busy to engage with tool	Phased rollout — start with 1 company, expand after 2 weeks	Medium

What success looks like at 90 days: A clear, documented improvement in AI Readiness Scores across the organization's active report estate, at least one analyst describing the tool as 'part of their workflow,' and enough evidence to credibly demo the tool to an external Power BI shop.